GitHub: From Code to Collaboration

Your code. Saved, tracked, and shared.

GitHub is a platform where developers store, manage, and collaborate on code.

Store your projects online.

Track every change you make.

Collaborate with developers anywhere.



Key takeaway: Think of GitHub as Google Drive for code, but with a complete history of every change.

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Git vs GitHub

Two different tools that work together.



Key takeaway: Git tracks your work. GitHub connects your work to the world.

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How GitHub Works

Understanding the basic workflow.

01 **Edit**
Write code locally

02 **Commit**
Save a snapshot

03 **Push**
Upload to GitHub



Pull brings remote changes back to your computer

Three terms to remember:

Repository (Repo):
Your project and its history.

Commit:
A saved checkpoint of your changes.

Push / Pull:
Send changes to GitHub or receive them.

Key takeaway: Your code lives locally, while GitHub keeps an online copy you can synchronize.

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Getting Started

Set up GitHub in three steps.

01



1. Create an account:
Sign up at github.com.

02



2. Install Git:
Download Git from git-scm.com.

03



3. Create a repository:
Click New repository, enter a name, and create it.

Your first repository is now ready to hold your code.

```
Example repository:
my-first-project
```

Key takeaway: A repository is the starting point for every GitHub project.

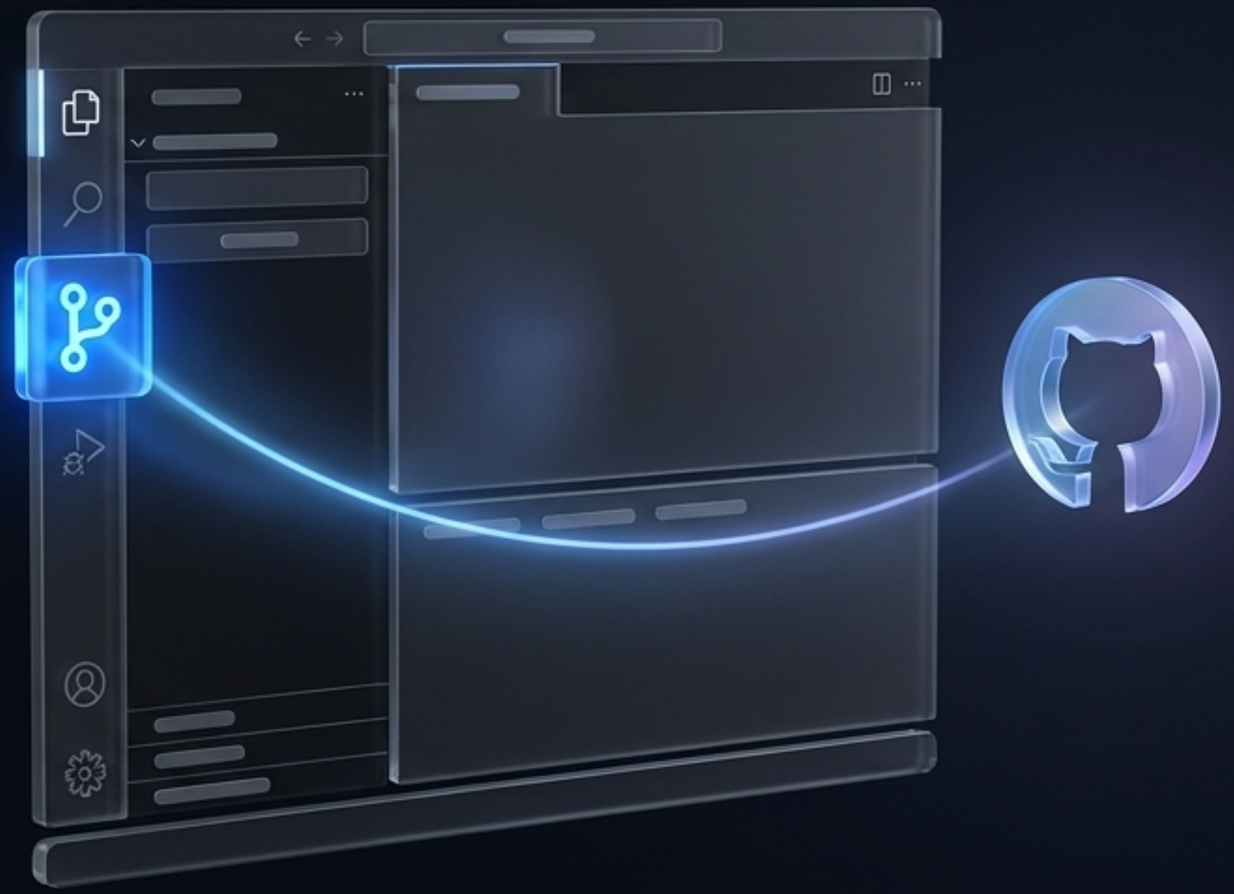
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Connecting GitHub to VS Code

Manage your code without leaving your editor.

1. Open Visual Studio Code and sign in to GitHub.
2. Open your project folder.
3. Click **Source Control** in the sidebar.
4. Select **Initialize Repository** for a new local project.
5. Use **Publish to GitHub** to create and upload the remote repository.

Alternative: Use Git: Clone from the Command Palette to download an existing repository.



Key takeaway: VS Code lets you edit, commit, push, and pull from one place.

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Your First Commit

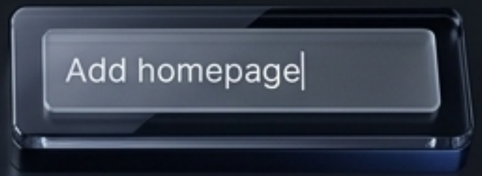
Save your first change to GitHub.

Make a small change to a file in VS Code. Then:

1. Open **Source Control** and review the changed files.
2. Click **+** to stage your changes.

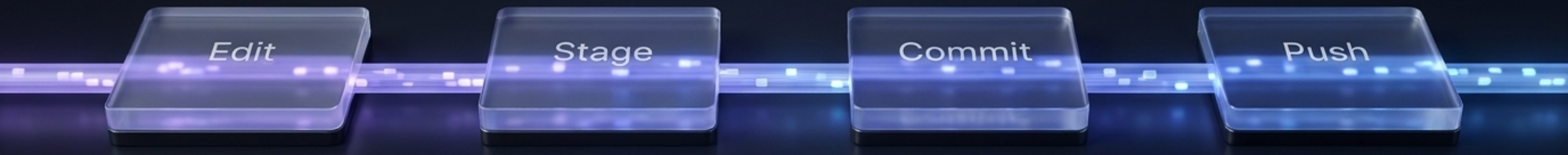


3. Write a commit message, such as **Add homepage**.



4. Click **Commit**, then **Sync Changes** or **Push**.

THE WORKFLOW

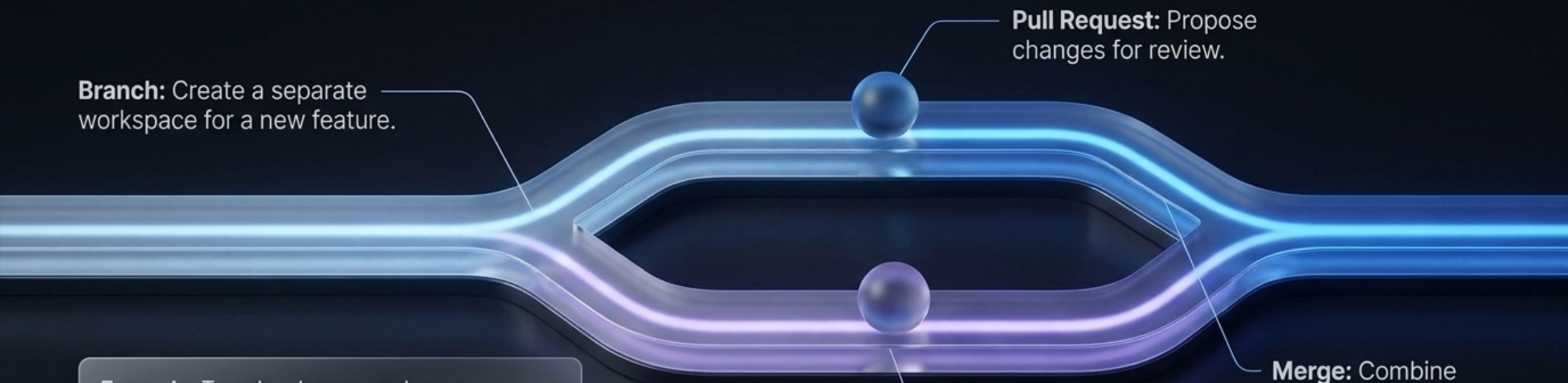


Key takeaway: A commit saves a version locally. A push sends it to GitHub.

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Working Together

Build as a team without overwriting each other's work.



Example: Two developers work on a website. One improves the homepage while the other builds the login page. Both changes can be merged into the same project.

Key takeaway: Branches let multiple developers work independently while keeping the main project stable.

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Ready to Build?

Everything you need to remember.

Repository
Stores your project

Push / Pull
Synchronizes your code

Commit
Saves a checkpoint

Branch
Separates new work



Your first challenge: Create a repository, connect it to VS Code, add a README.md file, commit your changes, and push them to GitHub.

Closing line: Build something. Save your progress. Share it with the world.

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